

SHREYAS S JOSHI

Bangalore, India | shreyasjoshi2511@gmail.com | +91 7892055781 | github.com/BlackPool25 | [linkedin.com/in/shreyas-s-joshi](https://www.linkedin.com/in/shreyas-s-joshi) | portfolio.shreyasjoshi.gzz.io | [LeetCode: BlackPool25 \(257\)](https://leetcode.com/BlackPool25)

SUMMARY

Pre-final B.Tech AIML (BMSIT, CGPA 9.36) + B.Sc CS (BITS Pilani Online, 8.91) — Expected May 2028. Backend/Systems Engineer focused on Dockerized services and protocol design. Built offline RaptorQ transfer protocol (QRStream, 170 commits, ~46KB/s measured) and 8-hour Docker-isolated runtime generator (BUILD & CONQUER 3.0 Winner, 1st Place). Python-first: FastAPI/Spring Boot, PostgreSQL/Redis, Docker/Linux.

SKILLS

Proficient: Python, Java, SQL — daily driver for backend services & DSA

Intermediate: C++ (DSA — 257 LeetCode: 132 Easy / 118 Medium / 7 Hard, 80% accept), C

Familiar: TypeScript, JavaScript, Dart — Flutter/PWA (QRStream interop), Solidity

Backend & Infra (Applied): FastAPI, Spring Boot, REST, WebSocket, BullMQ (Job Queue), Redis, Kafka (event streaming, Familiar), PostgreSQL (pgvector / PostGIS / pgRouting), ChromaDB, Docker, Docker Compose, GitHub Actions CI, Linux

Systems & Protocols (Applied): RaptorQ fountain code, CRC-32C / SHA-256, event-driven async (Spring ApplicationEvents + @Async), pgRouting Dijkstra, stateful map-matching, WebSocket streaming, Flutter Rust Bridge FFI

PROJECTS

QRStream — Offline Erasure-Coded File Transfer Protocol | github.com/BlackPool25/QRStream | Python sidecar + Dart/Flutter + TypeScript | 170 commits, MIT, CI v1.0.1

- Designed wire-compatible offline protocol: RaptorQ fountain-encode K symbols → cycling QR grid (1×1 to 3×3, V27/V34/V40, 12–30fps, metadata every 32 ticks for late-join) → ZXing-C++ decode → FrameBuffer dedup by ESI → any K of K+repair symbols rebuild + SHA-256 gate; transfer cap 16 MiB.
- Implemented Rust codec via flutter_rust_bridge 2.12 FFI (71KB Rust + 27KB C++) with Python verification pipeline, continuous linear tile scaling (nearest-neighbour hard edges, camera-decodable), per-frame CRC-32C + whole-file SHA-256; honest throughput ~46 KB/s (1 MiB ~23s measured) via Playwright headless + virtual camera — docs/PERF.md.
- Shipped 3-OS releases: Linux install.sh (SHA256SUMS verified) + RPM, Android APK (stable keystore via GitHub Secrets — fixes INSTALL_FAILED_UPDATE_INCOMPATIBLE), Windows installer (MSVC runtime app-local); 118 Flutter tests; ARCHITECTURE + THREAT-MODEL documented.

AgentDock — Visual Builder → Docker-Isolated Agent Runtimes — BUILD & CONQUER 3.0 WINNER (1st Place, 8hr, Altreno Club BMSIT) | Team: Aditya Prasad, Varnika Kirani Raghavendra | *TypeScript, Bun, Hono, BullMQ/Redis, FastAPI/Python, Dockerode, React Flow*

- Built React Flow canvas that compiles natural-language agent descriptions into fully self-contained Docker Compose runtimes (API gateway + Redis + FastAPI workers) — no coupling to builder backend; judges stopped rubric to ask to run locally vs single-purpose apps.
- Solved container isolation under timer: volume mounts + network routes for dynamically spawned agents to share files / communicate; debugged network routes + volume mounts as timer ran down — achieved reliable isolation where other teams hardcoded single apps.
- Engineered orchestration: Hono gateway + BullMQ job queue (Redis), Python FastAPI agent runtime, RAG memory via ChromaDB/offline sentence-transformers (per-agent Git-backed markdown volumes), LLM Gateway (5 providers: OpenAI/Anthropic/Groq/Gemini/Ollama with failover + rate limiting).

ZonePilot — Fleet Zone Compliance Engine | github.com/BlackPool25/ZonePilot | Java 25, Spring Boot 3.5, PostgreSQL 16 + PostGIS + pgRouting, Flyway, Docker | 47 commits | Live: zonepilot-snowy.vercel.app

- Architected Spring Boot 3 fleet brain: pgRouting directed Dijkstra on Bangalore OSM (blr_2po_4pgr, 1000× zone penalty, 5-attempt recursive compliance + wait-state via Google Routes), Flyway migrations, Docker Compose + partitioned vehicle_position_log (monthly) + 5 GIST spatial indexes.
- Implemented atomic breach detection inside PostgreSQL trigger (trg_detect_zone_breach AFTER INSERT) — no missed violations on app crash — plus stateful map-matching (30m snap with heading, 50m off-route 2-tick reroute), IST timezone overnight curfew logic, ST_Covers boundary handling.

OPEN SOURCE

opentofu/opentofu#4328 (merged) — Fixed plan file serialization for ForgetThenCreate (lifecycle.destroy=false): bridged Go ↔ protobuf enum (Action_FORGET_THEN_CREATE=10) in internal/plans/planfile/tfplan.go (+7) + tests (+25) — github.com/opentofu/opentofu/pull/4328

ACHIEVEMENTS

- 1st Place — BUILD & CONQUER 3.0 (Altreno Club, BMSIT) — 8-hour prototype, AgentDock Docker-isolated runtime generator.
- 2nd Place — CTF Contest (Access Denied Club, BMSIT) — competitive capture-the-flag placement.
- DSA: 257 LeetCode solved (132 Easy / 118 Medium / 7 Hard) — 80.06% acceptance, beats 93.73% runtime — C++ primary for DSA, Python for backend prototyping — github.com/BlackPool25.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning — BMS Institute of Technology and Management, Bangalore | CGPA 9.36 / 10 | Expected May 2028

B.Sc, Computer Science — BITS Pilani (Online) | CGPA 8.91 / 10 | Expected May 2028